

Modelling automation. A mix-method exploration.

Introduction and literature review

Automation has become one of the central economic issues due to rapid advancements of artificial intelligence, robotics, and machine learning across a wide spectrum of industries (Hutson & Ceballos, 2023). While earlier technologies automated routine manual or clerical tasks, modern AI increasingly performs tasks associated with educated workers. Economists increasingly disagree about whether AI will primarily substitute for workers or complement them. Acemoglu & Restrepo (2019) argue that technological change has recently been biased towards automation rather than reinstatement. This matters as automation and reinstatement have divergent implications for labour demand, wages and inequality.

However, it is important to realise that technological advancements do not necessarily result in negative outcomes for labour. Autor (2024) argues that modern AI may function as a “inversion technology” that can restore the “middle-class” area of the labour market. Before advancements in LLMs computational progress was largely focused on automation of low-skilled workers and hugely favoured experts. However, as Autor highlights AI presents an opportunity for democratisation of those high skilled jobs, by providing real-time guardrails for workers without elite degrees who can start performing tasks that were previously reserved for engineers, lawyers and doctors. Whether AI is predicted to destroy jobs or create new opportunities depends heavily on how automation is represented in economic models.

Earlier macroeconomic studies modelled automation as factor augmenting (Nordhaus, 2015) primarily through the standard CES aggregate production function¹. The factor-augmenting modelling framework implies that an improvement in capital technology increases the economic output which in turn makes wages (also known as the marginal product of labour) rise (see Appendix B figure 1). While wages rise regardless, the labour share is dependent on whether the elasticity of substitution between capital and labour (ε) is less or greater than one. Most literature places the elasticity coefficient between 0.5 and 1 which means that further technological progress would not reduce the demand for labour and strictly increases the equilibrium wages (Acemoglu & Restrepo, 2018).

Labour-augmenting technology works in a similar way. For realistic parameter values i.e. $\varepsilon \in (0.5, 1)$ the labour-augmenting technology always increases labour demand and equilibrium wage (Acemoglu & Restrepo, 2018). However, this type of technology decreases the labour share for these elasticity values (see Appendix B figure 1). This approach creates a frictional tension between capital- and labour-augmenting technologies. However, unless the goal is to examine this specific tension the other predictions are not intuitively appealing: for example, strict increase of labour demand disregards the automation substitution effect, wherein specific tasks, capital replaces labour entirely (Acemoglu & Restrepo, 2018). To

¹ Please note, that the same study later adopts a more cautious view of long-term economic impact and automation constraints (Nordhaus, 2021).

mitigate this, this study will model the automation effects through a task-based approach outlined by Acemoglu & Restrepo (2018).

Acemoglu & Restrepo propose modelling the aggregate output as combination of a continuous range of tasks indexed by $i \in [N - 1, N]$, where i corresponds to a specific task index and N is used to govern the upper and lower boundaries of the task space i.e. to exclude tasks that disappeared or tasks that have not been created yet. The automation threshold is defined as $I \in [N - 1, N]$ and corresponds to the marginal task that can be completed by a machine. Therefore, $i \leq I$ are automated tasks and are done by capital and $i > I$ are non-automated tasks done by labour. The automation process is modelled by expanding the I boundary. Machines expand into tasks previously reserved just for labour which is the *extensive margin of automation*. The increase of the efficiency of machines for already automated tasks $i \in [N - 1, I]$ is the *intensive margin of automation*².

Task-based model also relies on monetary principles meaning that a task is considered automated only if this task is cheaper to produce using capital:

$$\frac{\gamma(i)}{\eta(i)} < \frac{W}{R},$$

where W is wage and R is the rental rate of capital, and $\gamma(i), \eta(i)$ are their respective productivities (intensive margin). Autor et al. (2003) state that the declining price of computing power is the “casual force” behind the rapid adoption of computer technology. In summary: *it is not enough to invent a machine; a machine must be cost-effective*. However, while machines require an exponential increase in computing power to achieve linear gains in ability, historically Moore’s Law has provided a balanced improvement in cost (Thompson et al., 2023)³.

The total output is defined as a CES aggregate over individual task outputs⁴:

$$Y = \left(\int_{N-1}^N y(i)^{\frac{\sigma-1}{\sigma}} di \right)^{\frac{\sigma}{\sigma-1}}$$

Where $\sigma > 0$ is the substitution elasticity between different tasks.

Value	Meaning	Labour share
$\sigma \rightarrow \infty$	perfect substitutes	decreases
$\sigma \rightarrow 0$	perfect complements (Leontief)	increases
$\sigma = 1$	Cobb–Douglas benchmark	static

When tasks are great substitutes, it makes it easier for a machine to adapt the knowledge it learned from one task to another. Say there are two distinct tasks:

² Note, that this study examines specifically extensive automation.

³ However, recently Moore’s law has been breaking down which has profound implications for rapidity of AI development. Progress will depend more heavily on algorithmic intensive improvements rather than relying on hardware alone to overcome diminishing marginal returns of computational performance.

⁴ When calculating the aggregate using the automation threshold I definition we can integrate the capital and labour tasks separately (see function 7 from A&R paper).

- Checking fire alarm detector functionality.
- Troubleshooting Wi-Fi connectivity.

These tasks are distinct and done by different professions. But in σ terms they are treated as highly substitutable not because of the similarity of the domain, but because of the shared cognitive and physical structure. Both require diagnostic inference over partially invisible systems, coming up and testing a hypothesis and verification. The procedural knowledge required for the execution of these tasks is inherently different (one requires fire safety system knowledge, the other network knowledge), however both tasks are executed with a similar framework of programmable instructions. This means that if a machine can do one of these tasks and sufficient theoretical training data is available, said machine can adapt that knowledge to a different task⁵. This is the core principle that this study's modelling augmentation is based on. The model assumes that task allocation is determined primarily by relative productivity across factors, however many complex cognitive tasks may require a complimentary nature of machine capabilities rather than substitution based on advantage.

This leads to the reframing of the automation threshold that this study investigates. Instead of modelling AI development as a 1-dimensional expansion of the automation threshold I , this study proposes expanding automation into a 2-dimensional plane and separating the physical and cognitive complexity of a given task. This is motivated by the difference between automation bottlenecks facing algorithmic machines from those facing robotics. Frey & Osborne (2017) differentiate perception and manipulation from creative intelligence. For example, janitorial services are low in cognitive complexity, however the physical aspect is where many modern solutions struggle as physical environment is much more unpredictable. It is possible to extend the capability of a cognitive machine through extensive scaling of computing power. On the other hand, extension of robot capability in the physical domain requires intensive restructuring – a completely different approach.

Moreover, Acemoglu & Restrepo (2018) impose a hard boundary condition where a task is either done by a machine or labour. This study views it as inconsistent with modern AI adoption where a machine can be integrated with human supervision first (a co-pilot) and only later as it becomes more proficient can overtake other parts of a task. This changes how we view automation from binary “Either capital or labour” to “Either capital or labour OR both”. To satisfy these conditions this study uses a diffusion-based model to estimate machine development. However, quantitative modelling alone cannot locate real occupational tasks on the complexity surface, to bridge this gap this study used a complex sequential mix-methods design (Plano Clark, 2016) in which an initial replication and reframing of a task-based model (Acemoglu & Restrepo, 2018) is followed by a qualitative close reading section that developed a structured questionnaire framework for evaluating task-based complexity. The qualitative framework is aimed to quantify a sample of qualitative data: task descriptions available in the O*NET database and is designed to support future empirical calibration and model refinement.

⁵ A machine learning cross-domain methodology application.

Application of methods

2. Quantitative method application

1. Domain definition

We define a 2D grid T where $i, j \in [0, 1] \times [0, 1]$. Instead of ordering tasks on a single 1D automation plane I , each task is defined by two coordinates (i, j) where:

- $i \in [0, 1]$ represents the **Cognitive Complexity** (0 = simple logic, 1 = abstract reasoning/creativity);
- $j \in [0, 1]$ represents **Physical Complexity** (0 = repetitive motion, 1 = intricate motion, hard labour).

The space is normalised to a unit square as it simplifies the aggregation and can also be interpreted intuitively as a decimal form of percentages. In other words, $j = 0.6$ means that machines have mastered a task located at 60% of the maximum complexity of physical labour.

2. Productivity structure

This modelling approach assumes a comparative advantage of labour in high cognitive $i \rightarrow 1$ and high physical $j \rightarrow 1$ tasks, while machines have the advantage in tasks which are low cognitive (e.g. routine calculations) and low physical (e.g. assembly line), but struggle as complexity increases in both directions. Moreover, since this study focuses specifically on extensive automation (for example, current AI tools expanding into more complex tasks without improving their intrinsic performance), the simulation does not alter the productivity values of either capital or labour, $\gamma(i, j)$ and $\eta(i, j)$, respectively. Task productivity is specified such that machine productivity $\eta(i, j) \rightarrow 0$ as $i, j \rightarrow 1$.

Note, that the productivity is clipped to 10^{-4} . This is numerically motivated to make sure there are no negative productivity values, however it can be interpreted structurally. No task can reach a full “non-automatable” state, instead most complex tasks retain a vanishing but non-zero probability of automation. Therefore, automation is governed by a continuous cost-productivity trade-offs rather than impossibility conditions. Therefore, this study defines complex tasks as theoretically reachable but economically and mathematically unlikely which could be a more rigorous modelling approach than trying to establish where the machine learning boundary lies.

3. Capability-cost condition

There is a conceptual tension in the reframing of the automation threshold. Acemoglu & Restrepo (2018) define I as a cost condition (i.e. it must be cheaper to use capital). However, my matrix represents machine ability from a purely extensive standpoint while abstracting technical capability from cost of task execution. In general, this mapping is seen as non-linear. Thompson et al. (2023) show that linear improvements in performance often require

exponential increase of computing input. That implies that capability might not translate into cost reduction.

4. Model specification

To satisfy assumptions automation is modelled as an expansion into both directions through a heat-diffusion process, where σ elasticity coefficient corresponds to the diffusion constant (commonly referred to as D). This is motivated by the assumption that replaceability between tasks correlates to how easy it is to reuse learned knowledge between one task and another.

However, the standard heat-diffusion equation proved incomplete as one of the defining features of the heat equation is constant ‘leakage’ of information from a specific grid point to its neighbours. This created an effect where machines gain ability in complex tasks while staying mediocre at easy ones. This can be interpreted as machines skipping the proceduralisation stage of knowledge acquisition process defined by Anderson (1982).

Following Anderson’s theory on skill acquisition, expertise develops through the proceduralisation stage where descriptive knowledge is transformed into executable processes through practise. This is why the diffusion of AI capability across related tasks is modelled using a Fisher-KPP reaction-diffusion process. These types of equations are widely used in mathematical biology to represent gradual propagation instead of an instant transition (Murray, 2003)⁶.

$$\frac{\partial T}{\partial t} = \sigma \nabla^2 T + \gamma T(1 - T)$$

Where the components are:

- **Diffusion** $\sigma \nabla^2 T$: knowledge “leaks” to other tasks.
- **Learning** $\gamma T(1 - T)$: once machine learns a little bit about a specific task, it starts using that knowledge to rapidly train itself (local growth).

5. Numerical scheme

To solve this equation the study uses a numerical finite-difference scheme.

For the diffusion component we get:

$$U_{i,j,n+1} \approx r(U_{i+1,j,n} + U_{i-1,j,n} + U_{i,j+1,n} + U_{i,j-1,n}) + (1 - 4r)U_{i,j,n}$$

where $r = \frac{\sigma \Delta t}{h^2}$.⁷

For the learning term⁸:

$$\gamma T(1 - T) \approx \gamma U_{i,j,n}(1 - U_{i,j,n})\Delta t$$

⁶ An interactive simulation is available on this study’s GitHub repository.

⁷ For more information on how to solve the heat equation, you can refer to my previous work of modelling the heat equation on a circular domain (Kolisnyk, 2025).

⁸ We must multiply it by Δt to ensure consistency with the diffusion term in relation to r .

Combining the two parts we get:

$$U_{i,j,n+1} \approx r(U_{i+1,j,n} + U_{i-1,j,n} + U_{i,j+1,n} + U_{i,j-1,n}) + (1 - 4r)U_{i,j,n} + \gamma U_{i,j,n}(1 - U_{i,j,n})\Delta t$$

With $r = \frac{\sigma \Delta t}{h^2}$.

The first two terms are weighting averaging (standard heat equation components) which define how does knowledge spread out to different tasks. Whereas the last term corresponds to local non-linear learning.

To ensure numerical stability for the explicit-finite difference scheme the diffusion parameter must satisfy $r \leq \frac{1}{4}$, a standard condition for 2D schemes. Moreover, since the spatial step size h is implicitly uniformed into the task space, the stability condition is restated in terms of σ rather than $\frac{\sigma \Delta t}{h^2}$. In practise, r is scaled to $0.24 \cdot \left(\frac{\sigma}{\sigma_{max}}\right)$ which keeps it strictly below the 0.25 Von Neumann threshold for all values of σ .

Similarly, the $\gamma \Delta t$ is capped at 0.05 to ensure that the non-linear term does not destabilise the model. $\gamma T(1 - T)$ reaches its maximum learning rate at $T = 0.5$, where it equals 0.25γ . Therefore, the maximum step size learning contribution is 0.0125 which is well within the $[0, 1]$ range.

Both values are hard coded to ensure stability across all tasks and time steps.

6. Integration into CES

Fisher-KPP simulation produces a $T_{i,j}$ matrix for every time step h^9 . This matrix feeds directly into a CES production framework. Each value of $T_{i,j} \in [0,1]$ represents the amount of automation of that task, which changes over time. A value of 0 represents that a task is fully done by labour and a value of 1 means that it is fully automated. The human labour share of a given task is $1 - T_{i,j}$.

The model integrates across all tasks to produce aggregates of capital and labour – A_K and A_L , which enter the CES production function to produce output Y , wages W and labour share s_L . This is repeated for every snapshot of T and for every value of $\sigma \in [0.2, 3]$. So, the outcome are 3D plots of these 3 different time series for each value of substitution elasticity.

3. Qualitative method application

This study's methodology runs sequentially where this qualitative section proposes a framework on how to create usable data to later use in the reframed task-based model this study investigates. The goal of this section is to estimate physical and cognitive complexity for each task for 2 different jobs: firefighter and paediatric surgeon. These occupations were selected as while both requiring physical and cognitive capability, they exist on the opposite

⁹ Note that the code uses specific snapshots for performance to feed into the CES production function.

ends of the complexity grid where most firefighter's tasks are highly physical while paediatric surgeon focuses more on the cognitive aspect of task complexity.

1. Data source and selection

The data source for the close reading analysis is a sample of the Occupational Information Network (O*NET) database. O*NET is an online database developed under sponsorship of the United States Department of Labor which is one of the primary sources of occupational information in the world. O*NET was selected due its uniquely standardised task-level descriptions, which are not available in equivalent UK occupational datasets.

Specifically, this study uses 2 O*NET datasets:

- **“Task Statements”** which lists all tasks that employees do for a specific job title;
- **“Tasks to DWA”** which lists DWAs (Detailed Work Attributes) for every task.

DWAs are particularly useful as they bring an additional layer of nuance. For example, paediatric surgeons have a task: “Manage surgery services, including planning, scheduling and coordination, determination of procedures, or procurement of supplies and equipment”. DWAs split it into several work attributes:

- Manage healthcare operations;
- Order medical supplies or equipment;
- Schedule patient procedures or appointments.

These attributes inform a more nuanced understanding of task complexity.

2. Close reading methodology

Close reading is a form of systematic attentiveness in which the reader investigates specific features of a text in detail to understand how it produces meaning (Ohrvik, 2024). Within this study, close reading is operationalised as a questionnaire, an instrument through which each task and DWA attributes are read systematically (see Appendix C for questions, criteria are discussed in the questionnaire section) against ten theory-grounded criteria.

For each of the ten questionnaire criteria (5 physical, 5 cognitive), a score of 0, 0.5, 1 is assigned. The cognitive and physical coordinates is computed as a mean of its 5 respective criteria. These questionnaire constraints will produce a score for the complexity of a given task independently for i, j (cognitive and physical complexity).

Following the scoring, all tasks for each occupation are mapped on the complexity grid (i, j). The outcome distributions are evaluated against prior theoretical expectation. Paediatric surgeon is a blueprint of a profession that requires both physical and cognitive dexterity, while a firefighter is more physical and requires work in high-risk unpredictable environments. Therefore, the hypothesis that was tested is that paediatric surgeon's distributions will be heavily weighted towards high complexity (1, 1) grid point, while firefighters should be lean more towards high physical, medium cognitive (0.5, 1) grid point.

3. The questionnaire

The cognitive questions draw richly from the cognitive psychology literature on expertise (Hoffman, 1998), skill acquisition (Anderson et al., 1982 and Dreyfus & Dreyfus, 1986), high-performance or complex skills (Schneider, 1985) and education regarding technological change (Hutson & Ceballos, 2023). On the other hand, physical complexity is discussed less in the cognitive psychology discipline. Therefore, the physical questions lean more on labour economics studies on distinction between routine and non-routine tasks (Autor et al., 2003 and Levy & Murnane, 2004) and draws insights on algorithmic difficulty of physical automation discussed by Thompson et al. (2023).

The central piece of this qualitative methodology is the Frey & Osborne (2017) paper which focused on identifying specific machine-learning bottlenecks that currently limit machine's capabilities. These include:

- *Perception and Manipulation*. Tasks requiring manual labour and ability to work in unpredictable and unstructured workspaces.
- *Creative Intelligence*. Tasks involving originality.
- *Social Intelligence*. Tasks requiring social perception, negotiation or collaboration.

The Social Intelligence tasks are interesting in relation to skill. Green (2011) argues that skill is inherently a social concept and the definition of what is skilful and by extension what is complex is inherently as social construct.

By assessing these bottlenecks and integrating with multidisciplinary literature this study aimed to create a qualitative framework through which different tasks were mapped on the 2D complexity plane. The criteria are as follows:

Cognitive complexity (i)

1. Programmability.

Grounded in Autor et al. (2003) programmability criterion where 'routine' tasks are those that can be specified by a specific algorithm or instructions. Polanyi's paradox is an important concept: if we know more than we can tell, codification fails. However, Frey & Osborne (2017) shows that computerisation is no longer confined to rule-based software and automation is rapidly expanding even into tasks that require subtle judgement.

2. Tacit knowledge.

Grounded in 3 converging sources. Hoffman (1998) perceptual expertise and tacit knowledge paradox where experts deepen their pattern recognition skill to a point where they start 'seeing the invisible' and can become less able to verbalise their reasoning as they become more masterful. Green (2011) shows that there is a paradox of ease where the most complex tasks can appear effortless because the complexity has become unconscious. This is also supported by Dreyfus & Dreyfus (1986) who define that expertise *is* situated coping¹⁰.

¹⁰ There is an ongoing debate within cognitive psychology on what expertise is. Dreyfus is contradicted by Anderson's (1982) ACT-R framework which argues that being an expert is not a special characteristic, but an ability to maneuver a system.

3. Concurrent cognitive demand.

Grounded in Anderson (1982) where the cognitive memory load is the defining feature of the declarative (aka descriptive) stage of skill acquisition and problem solving. This question is created to evaluate how much theoretical knowledge (facts, history, rules) is being used. It is also supported by Schneider (1985) where high-performance skills require concurrent sub-task management under stress. Moreover, as mentioned previously a linear improvement at high levels requires exponentially more computation (Thompson et al., 2023), which suggests that tasks requiring peak human multi-stream processing are inherently complex.

4. Social and emotional inference.

Grounded in Deming's (2017) *Theory of the Mind* which is a psychological concept of being able to put oneself in another person's shoes. This theory is aligned with the 'Social Intelligence' bottleneck described by Frey & Osborne (2017) and is data-driven as Deming (2017) shows that the labour market has seen a significant shift towards occupations that require social skills. Green's (2011) interdisciplinary synthesis of skill contributes a sociological perspective that emotional and aesthetic labour were historically 'invisible' because of the tacit nature of the tasks. These tasks remain among the hardest to automate as they are inherently contextual and as far as this study is aware there is no formal unified theory that can reduce this knowledge to 'trainable data'.

5. Cross-domain synthesis.

Grounded in Hutson & Ceballos (2023) who argue that during rapid technological change the most durable tasks are the ones requiring integration of multiple domains and application of abstract concepts in novel ways.

Physical complexity (*j*)

1. Environmental variability.

Grounded in Frey & Osborne (2017) perception and manipulation bottleneck and Autor et al. (2003) distinction that non-routine manual tasks require constant adaptation to an uncontrolled chaotic environment. It also supported by Hoffman (1998) training bottleneck where some physical scenarios are extremely rare, and it may take years before a novice encounters a particular scenario which makes it impossible to prepare for them.

2. Feedback dependence.

Grounded in Frey & Osborne (2017) perception and manipulation bottleneck and Green's (2011) worker autonomy concept. In tasks that require sensory-adjusted physical discretion external guidance may fail. In these tasks the feedback loop can be internal and invisible to the supervisor or specifiable to a machine.

3. Stimulus response variability.

Grounded in Schneider (1985) 'Practise makes perfect' fallacy. Tasks which consistently produce similar responses and portray similar information can be perfected through

repetition. When stimulus response varies it disrupts the proceduralisation process also known as the imperative stage of skill acquisition (Anderson, 1982). This internalised pattern recognition process will fail if the rule itself must change on context which keeps the task permanently complex.

4. Precision and Risk.

Grounded in Thompson et al. (2023) exponential scaling finding of the computational cost and Frey & Osborne (2017) perception and manipulation bottleneck. However, tools like the da Vinci Surgical System exist but they are not autonomous yet, because of the extent potential risk of a malfunction.

5. Concurrent Motor Independence.

This criterion was created as a physical equivalent of the Cross-Domain Synthesis. It is also grounded in Anderson (1982) finding that cognitive load degrades motor performance when both are demanded concurrently.

Outputs

The i, j coordinates from the qualitative section place real tasks on the same grid the Fisher-KPP model uses. This mixed-mixed method approach produces distributions that are displayed as a comparative plot (see Appendix B figure 1). The plot does not fully match the distribution hypothesis for tasks done by a paediatric surgeon where a paediatric surgeon's physical complexity distributes to a mean range of $j \in (0, 0.6)$ which seriously underrepresents the physical complexity of surgery. This highlights an important limitation of overall job complexity assessment – surgeons have a limited number of physical tasks which drags the mean complexity value down. On the other hand, this by itself could be a preliminary finding: surgeons do not always work on extremely complex tasks, the majority of tasks are analytical or procedural. This shows that specific tasks within paediatric surgery (and by extension other high-skill jobs) can be partially automated. This can be partially aligned to Autor's (2024) claim that automation can democratise high expertise jobs.

Quantitative modelling shows that both wages and total output rise in the short term (graphs are available in this study's GitHub repository) as machines take over routine programmable tasks where they exhibit higher level productivity compared to labour. That said, as the model represents an extensive expansion of automation, the productivity constraints negatively impact these metrics over the long-term.

This correlates to automation's impact on labour share. In lower σ values automation has little to no impact on the labour share followed by a steep decline which represents machine's ability expanding towards the whole complexity grid.

Moreover, the model behaves differently as when $\sigma \rightarrow 1$ the CES function converges to the Cobb-Douglas production function where the algebraic manipulation breaks, even though the true limit exists. While falling outside the scope of this study this special case can be

examined separately. This σ value means that firms can substitute capital and labour smoothly and proportionally which is the balanced flexibility case.

However, firefighter's tasks match the hypothesis distribution much more closely. This can highlight a hard labour bias within the questionnaire, however as highlighted previously can be empirically motivated as hard labour is a specific example of tasks that are automation resistant.

Critical reflection

This paper assumes that a task which is both physical and cognitive does not exhibit emergent complexity qualities. The overall complexity of the task is simply as a sum of the physical and cognitive aspects, but a case can be made that a task which requires both is inherently more complex and should be ranked higher.

When interpreting the questionnaire scoring outputs, three sources of systematic researcher bias must be acknowledged. Firstly, the researcher is by definition a non-expert coder, the scores are assigned speculatively without a first-hand experience with the tasks at hand. This creates a vulnerability specifically for the tacit knowledge criterion. Hoffman (1998) states that the most complex features of a task remain invisible to those who lack domain expertise, meaning that tacit knowledge scores will be systematically underestimated. Secondly, single-coder design is susceptible to human error and misunderstanding, therefore this framework does not claim complete objectivity, only transparency. Third, as Rinehart (1998, in Freeman, 2025) argues, the categories that researchers impose “*often tell us more about us, about our stance on how things are, than they do about any truth*” of the phenomena categorised.

Future research might focus on expanding the qualitative questionnaire to other professions. This can be done similarly using close reading and qualitative analysis, however this approach is resource heavy if used to rank all tasks from the database. This study proposes an integration of the questionnaire into a NLP model which has the ability to systematically look at language use and limit bias that arises from purely qualitative grading system.

Alternatively, the same grading can be done by a group of experts, a peer-review mechanism to mitigate the individual researcher bias.

Moreover, integration of intensive automation through empirical calibration of η and γ productivity values. This will also allow for a more complete picture of automation's impact on wages and total output across the economy. Future automation modelling must include task creation by shifting the unitary square boundaries and removing tasks that are no longer necessary.

Appendix A. References

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Appendix B. The computational notebook (Python)

The computational notebook is available at:

https://github.com/Doctor2007/task_based_model

Specific plots that are directly referenced in this paper are attached below.

Figure 1. *Wages and labour share comparison for 2 distinct σ values.*

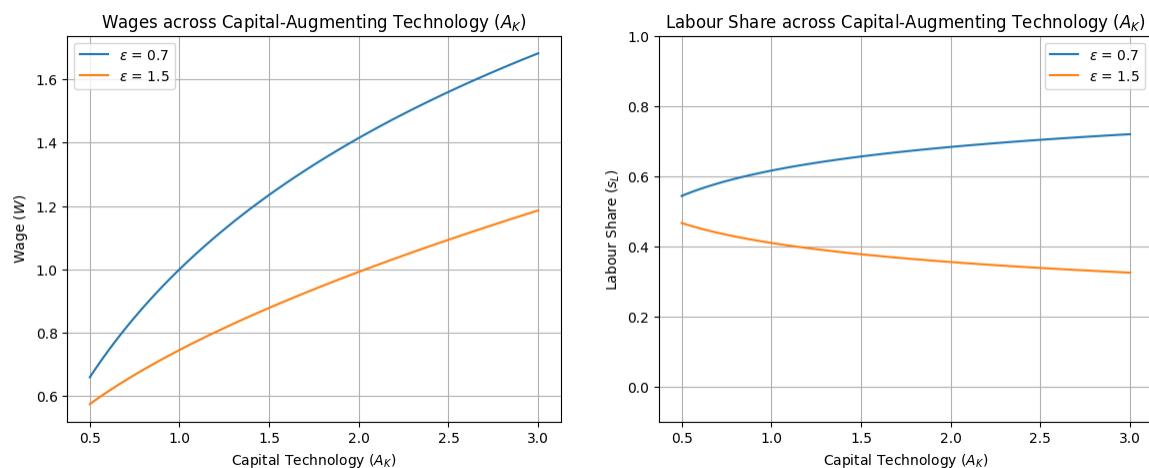
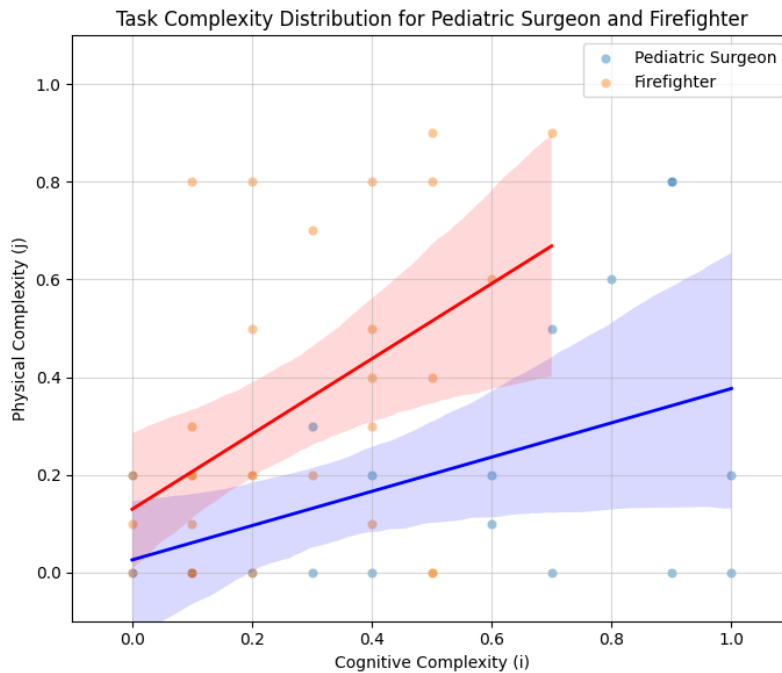


Figure 2. *Task complexity distribution comparison of a paediatric surgeon and a firefighter.*



Appendix C. Questionnaire

Cognitive complexity (i)

1. Programmability. *Does this task not reduce to fixed if $\rightarrow$ then statements or standard protocols in a way that a person without prior experience would be unable to execute it adequately?*
2. Tacit knowledge. *Does the expert rely on intuitive recognition of patterns or anomalies in a way where the expert may detect those features but cannot fully articulate?*
3. Concurrent cognitive demand. *Does this task require managing several streams of information where failure in one stream degrades the overall performance?*
4. Social and emotional inference. *Does this task require inference of another person's social and emotional states in real time in a way which varies and cannot be standardised?*
5. Cross-domain synthesis. *Does this task require integrating knowledge from various disciplines or transferring abstract concepts to genuinely novel situations where no existing templates exist?*

Physical complexity (j)

1. Environmental variability. *Does the physical execution of this task occur in an environment which changes unpredictably in real time, requiring constant adjustments that cannot be defined prior?*
2. Feedback dependence. *Does physical execution of this task depend on constant sensory feedback in a way where the removal of this sensory loop would substantially degrade performance?*

3. Stimulus response variability. *Does the same physical stimulus require different motor responses depending on context in a way which cannot be standardised and perfected through practise and repetition?*

4. Precision and Risk. *Does this task require extreme motor precision and continuous force calibration where a small deviation causes significant and potentially irreversible consequences?*

5. Concurrent Motor Independence. *Does this task require simultaneously controlling different limbs which perform different actions under sustained physical fatigue or high cognitive load?*

Appendix D. Questionnaire derived datasets

Table 1. Task complexity of a paediatric surgeon.

Task	P	TK	CCD	SEM	CDS	EV	FD	SRV	PR	CMI	i	j
0	0	1	1	0	0	0	0	0	0	0	0.4	0
1	0.5	1	1	0.5	1	1	0.5	0.5	0.5	0.5	0.8	0.6
2	0	0	0	1	1	0	0.5	0.5	0	0	0.4	0.2
3	0	0.5	0	1	0	0	1	0	0.5	0	0.3	0.3
4	1	1	1	1	1	1	0	0	0	0	1	0.2
5	1	1	1	1	0.5	0	0	0	0	0	0.9	0
6	0	0	0	0	0	0.5	0.5	0	0	0	0	0.2
7	0.5	1	1	1	0	0	1	1	0.5	0	0.7	0.5
8	0	0	0	0	0	0	0	0	0	0	0	0
9	0.5	0	0	0.5	0	0	0	0	0	0	0.2	0
10	0.5	1	1	0	1	0	0	0	0	0	0.7	0
11	0.5	0.5	0.5	1	0.5	0.5	0	0	0	0	0.6	0.1
12	0	1	1	1	0	0	1	0	0	0	0.6	0.2
13	1	1	1	1	0.5	0	1	1	1	1	0.9	0.8
14	1	1	1	1	0.5	0	1	1	1	1	0.9	0.8
15	0	0	0.5	0	0	0	0	0	0	0	0.1	0
16	1	1	1	1	1	0	0	0	0	0	1	0
17	0.5	0	0	0.5	0.5	0	0	0	0	0	0.3	0

Table 2. Task complexity of a firefighter.

Task	P	TK	CCD	SEM	CDS	EV	FD	SRV	PR	CMI	i	j
0	0.5	0.5	1	0.5	0	1	1	0.5	1	1	0.5	0.9
1	0	0.5	1	0.5	0	0	0	0	0.5	0	0.4	0.1
2	0	0	0.5	0	0	0	0	0	0	0	0.1	0
3	0.5	0	0.5	1	0	0.5	1	0	0	0	0.4	0.3
4	0.5	0.5	0.5	1	0	0.5	1	0.5	0	0	0.5	0.4
5	0	0	0.5	0	0	1	1	0.5	0.5	1	0.1	0.8
6	0	0	0	0	0	0	0	0	0	0	0	0
7	1	0.5	0.5	0	0	1	1	0	1	1	0.4	0.8

8	0	0	0	1	0	1	1	1	0.5	0.5	0.2	0.8
9	0.5	0	0	1	1	0	0	0	0	0	0.5	0
10	0.5	0.5	0	0.5	0.5	1	0.5	1	0	0	0.4	0.5
11	0.5	1	0	0	0	1	1	1	0	0.5	0.3	0.7
12	0	0	0	1	0	0	0	0	0	0	0.2	0
13	0	0	0.5	0	0	0	0	0	0	0	0.1	0
14	0.5	0.5	1	1	0.5	1	1	0.5	1	1	0.7	0.9
15	0	0	0	0	0	0.5	0	0	0.5	0	0	0.2
16	0	0.5	1	0.5	0.5	1	1	0.5	0.5	1	0.5	0.8
17	0	0	0	1	0	0.5	0	0	0	0.5	0.2	0.2
18	0	0	0	1	0	0.5	0	0	0	0.5	0.2	0.2
19	0.5	0.5	0.5	0	0	1	0	0	0	0	0.3	0.2
20	0	0	0	0.5	0	0.5	0	0	0	0.5	0.1	0.2
21	0	0	0.5	0	0	0	0	0	0	0	0.1	0
22	0.5	0	0.5	0	0	1	0.5	0	0.5	0.5	0.2	0.5
23	0.5	0.5	0.5	1	0.5	1	0.5	0.5	0.5	0.5	0.6	0.6
24	0	0	0	0.5	0	0.5	0	0	0	0	0.1	0.1
25	0	0	0	0	0	0.5	0	0	0	0	0	0.1
26	0	0.5	0	1	0.5	1	0.5	0.5	0	0	0.4	0.4
27	0	0	0.5	0	0	0.5	0	0	0	0.5	0.1	0.2
28	0	0	0.5	0	0	0.5	0	0	0.5	0.5	0.1	0.3
29	0.5	0.5	0	1	0.5	0	0	0	0	0	0.5	0

Appendix E. AI Declaration

In completing this assessment, generative AI tools (Claude, Anthropic) were used in:

- coding assistance in the development and debugging of the Python computational notebook, including implementation of the Fisher-KPP finite difference scheme and CES production function, particularly the vectorised logic through slicing to save computational time as for loops are extremely inefficient;
- explanation of complex mathematical and economic concepts, including reaction-diffusion equations and task-based modelling frameworks, to support the author's own understanding;
- initial scoping of relevant literature and research directions.

No AI generated text appears in this paper.